Jitter simulation and basic analysis

HW12 - Introduction to Signal Integrity and Electromagnetic Compatibility

This report recomputes all four required 40 Gb/s NRZ experiments using the supplied two-port channel. The transmitter contains no jitter, sinusoidal jitter (SJ), random jitter (RJ), or both. Receiver eye diagrams, one-UI transmitter and receiver jitter distributions, and their variances are obtained from the same channel and common random data. Every numerical value below comes from the saved run.

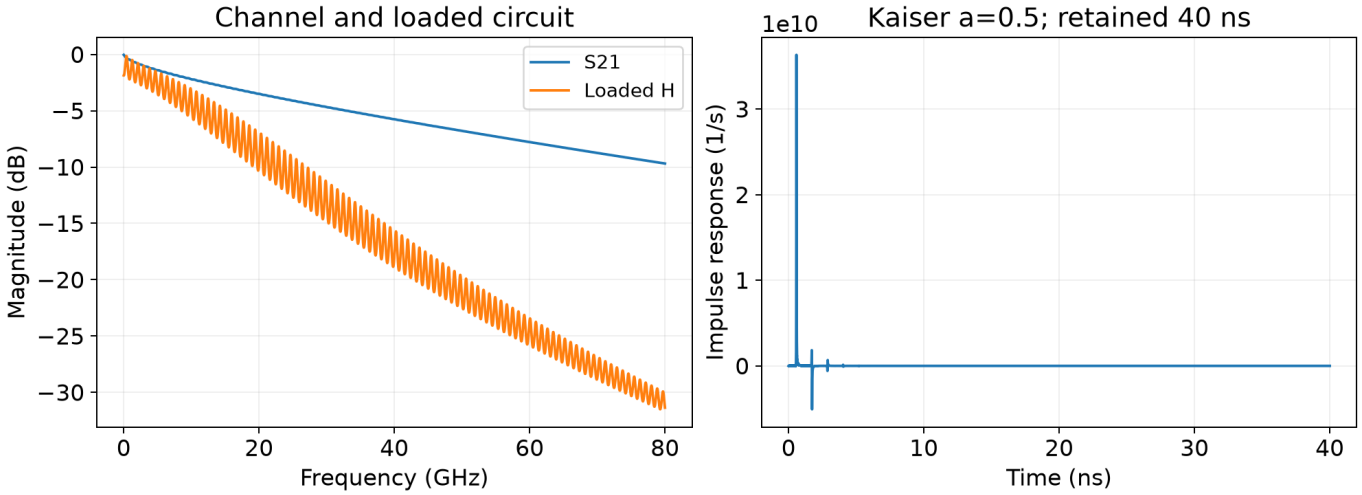
Physical and numerical model

Quantity	Setting
Bit rate / unit interval	40 Gb/s NRZ / 25 ps
Source voltage levels	0 and 1 V
Source resistance / capacitance	20 ohm / 0.3 pF
Load resistance / capacitance	85 ohm / 0.3 pF
Rise and fall duration	5 ps
SJ	3 ps peak amplitude, 1 MHz
RJ	Zero-mean Gaussian, 1 ps standard deviation
Evaluated data	1,048,576 bits per case; seed 0
Excluded startup	1,602 warmup bits per case
Time samples	200 per UI; 0.125 ps spacing
Eye voltage bin	1 mV
Kaiser window / retained impulse	alpha = 0.5 / 40 ns

The SJ period is 1 microsecond, or 40,000 UI. The measured record spans 26.2144 SJ cycles. The source resistance, load resistance and shunt capacitances are included in the source-to-load transfer H(f); H is not simply the channel's S21. There is no receiver additive noise or clock recovery model.

Channel and waveform calculation

The Touchstone data use a 50 ohm reference impedance. At each frequency, modified nodal analysis solves the external circuit together with the two-port relation `(I-S)V = Z0(I+S)Iport`. The channel is oriented with S21 transferring excitation from port 1 to port 2. The Kaiser-weighted positive-frequency data are completed by conjugate symmetry to obtain a real impulse response on the simulation grid. The causal 40 ns interval is retained as a finite impulse response (FIR).

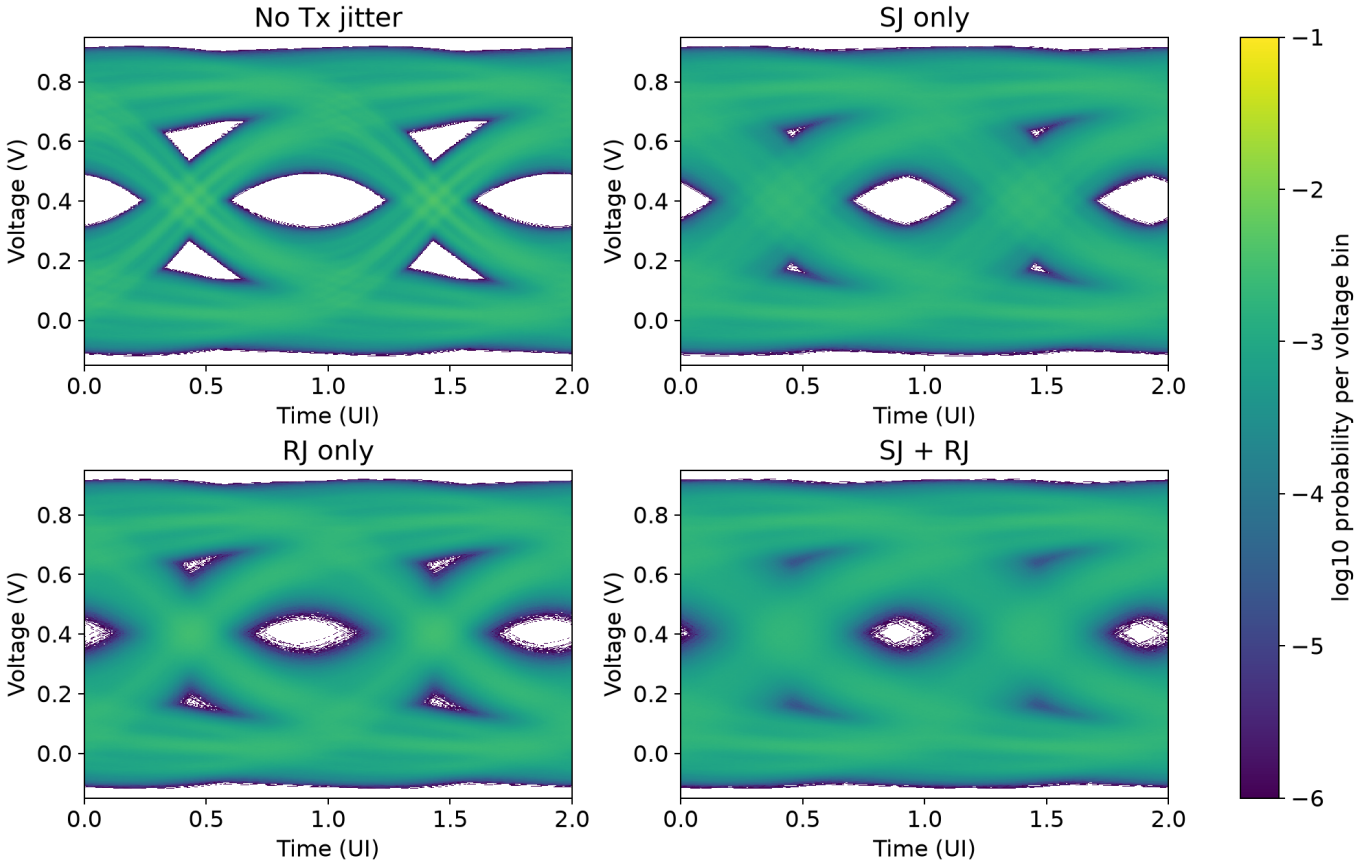


Loaded channel transfer and impulse response used in the four experiments. The original S21 is a reference; the source-to-load H includes both terminations and their shunt capacitances.

Channel diagnostic	Recorded value
frequency samples	8001
df hz	1e+07
maximum frequency hz	8e+10
reference impedance ohms	50
dc gain	0.806386
fir taps	320000
fir dc gain	0.804706
fir dc relative error	-0.00208306
fir last 1ns l1 fraction	9.60993e-05
s parameter orientation	S[receiving_port,incident_port]; no transpose

The retained FIR has DC gain 0.804706093, compared with 0.806385842 for the frequency-domain circuit. The signed relative difference is -0.2083%. The FIR is not rescaled to hide this windowing/causal-truncation difference. The receiver threshold remains the assignment's frequency-domain DC definition.

Four receiver statistical eyes



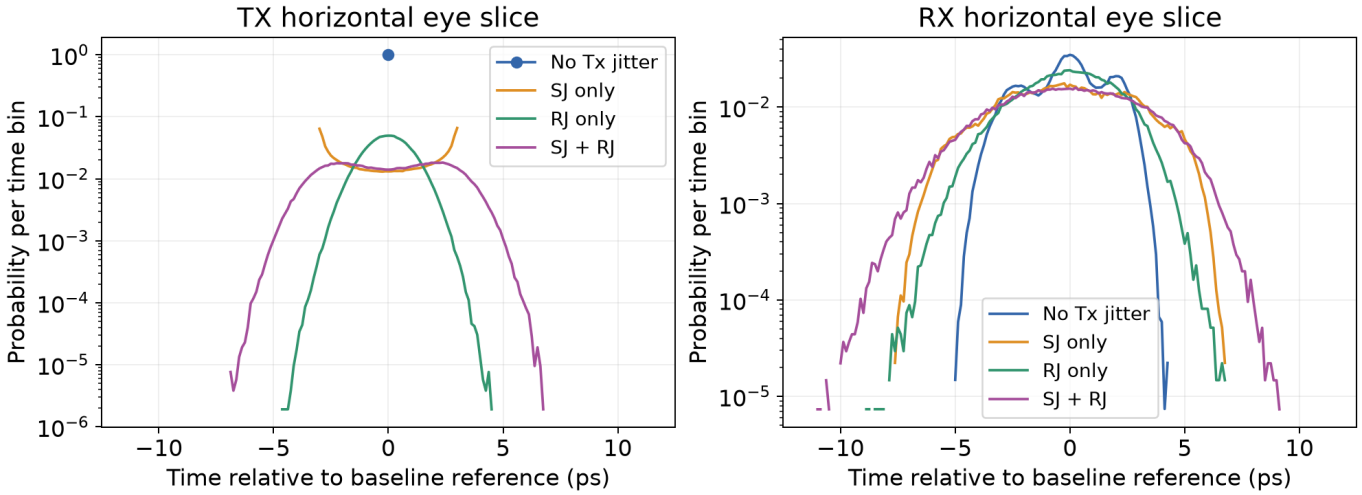
Receiver statistical eyes for cases 1-4. Every input includes the same finite rise and fall duration. Colors represent the per-time-column voltage probability; zero-probability bins are masked only for logarithmic display.

Case 1 already has receiver timing spread because channel memory produces data-dependent jitter (DDJ). The ideal transmitter does not remove channel ISI. Case 2 adds the slowly varying SJ displacement, case 3 adds independent random transition displacement, and case 4 combines both source impairments.

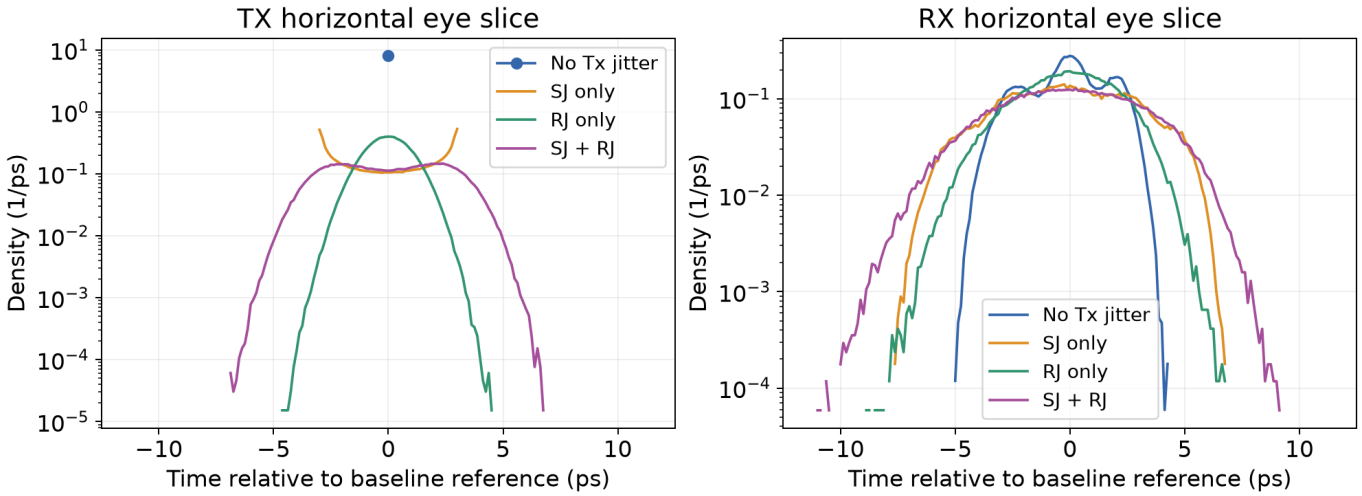
The simulation processes bounded blocks and carries the FIR state between them. Guard samples are excluded before accumulating exactly the requested number of payload bits. Every time column of each eye is normalized using its actual count. No artificial probability is added to the analysis arrays; zero-count bins remain zero. The two displayed UI repeat the same one-UI statistical eye.

Source edge displacements are rounded to the nearest time sample before waveform construction. The maximum timing-rounding error is 0.0625 ps. This matches the assignment's sampled-edge model and avoids the original truncation-toward-zero bias. It also prevents a narrow transmitter voltage slice from sparsely sampling continuous sub-grid edge positions. The rounding is a numerical approximation, not a physical bound on Gaussian RJ.

One-UI jitter distributions



Normalized transmitter and receiver jitter PMFs for all four cases. Each bin contains probability, and the probabilities sum to one.



The same distributions expressed as PDFs. Each PMF bin is divided by the time-bin width; density times the bin width sums to one.

The transmitter threshold is 0.5 V. The receiver threshold is half the loaded DC output, `real(H(0))/2`. A single one-UI interval is selected by circularly centering the baseline transition region; the reference is reused across cases at each observation side. This prevents losing bins when the transition straddles the phase origin. A PDF value has inverse-time units; a PMF bin is dimensionless.

The four receiver slices contain 134,840, 134,742, 135,727, 136,081 occupied voltage-bin samples, respectively. These counts, rather than the total transmitted bit count alone, set the smallest nonzero increments in the normalized slice PMFs.

Numerical statistics and transmitter theory

These are the assignment's **normalized eye-slice statistics**. The separate crossing-event diagnostic uses a different estimator and is identified explicitly.

Case	Tx variance (ps^2)	Rx variance (ps^2)	Tx RMS (ps)	Rx RMS (ps)
1	0	3.0074	0	1.73419
2	4.51963	7.53176	2.12594	2.7444
3	0.998228	4.28005	0.999114	2.06883
4	5.50517	8.77041	2.34631	2.96149

For samples at bin times t_i and a normalized PMF p_i, the statistics are

```
mean = sum(t_i * p_i)
variance = sum((t_i - mean)^2 * p_i)
centered RMS = sqrt(variance)
PDF_i = p_i / dt
```

The reported RMS is the standard deviation after removing the mean timing offset. Raw RMS would also include the square of a fixed delay or phase offset. Every time bin is spaced by the same dt used by the waveform simulator.

Tx case	Theoretical variance (ps^2)	Theoretical RMS (ps)
1: no jitter	0	0
2: SJ	4.5	2.12132
3: RJ	1	1
4: SJ + RJ	5.5	2.34521

For uniformly sampled sinusoidal phase, SJ has zero mean and variance A^2/2. Its density is `1/(pi*sqrt(A^2-j^2))` for `abs(j)<A`, with more probability near the turning points. Independent zero-mean Gaussian RJ has variance sigma_RJ^2. Thus the combined transmitter variance is `3^2/2 + 1^2 = 5.5 ps^2`, and its centered RMS is 2.345208 ps. Finite phase coverage, random sampling and time/voltage binning account for small measured differences from these limiting values.

Answers to the four assignment questions

Comparison	LHS (ps^2)	RHS (ps^2)	RHS - LHS (ps^2)	Residual / LHS (%)
Q1: Tx SJ + RJ	5.51786	5.50517	-0.012685	-0.22989
Q2: DDJ + Tx SJ	7.52703	7.53176	+0.00472376	+0.06276
Q3: DDJ + Tx RJ	4.00563	4.28005	+0.274415	+6.85074
Q4: intended Rx4	8.51258	8.77041	+0.257835	+3.02887

1. Does Tx case 2 variance plus Tx case 3 variance equal Tx case 4 variance?

Yes in the independent-source population model. For any two random timing errors, `Var(X+Y)=Var(X)+Var(Y)+2*Cov(X,Y)`. Here SJ phase and the Gaussian random source are independent, so their covariance is zero in the ensemble. The recorded comparison is 5.51786 versus 5.50517 ps^2, with a residual of -0.012685 ps^2. Finite-record and histogram differences need not be zero. Nearest-grid rounding also matters: rounding `SJ+RJ` does not always equal the sum of separately rounded SJ and RJ. Thus the sampled four-case experiment is not an exact algebraic sum even before finite-sample covariance is considered.

2. Does Rx case 1 variance plus Tx case 2 variance equal Rx case 2 variance?

Approximately, when SJ acts as a common slow time shift through the channel. A constant displacement of a waveform passes through an LTI channel as the same displacement. A 1 MHz SJ changes slowly relative to the 25 ps UI and the dominant channel memory, so its effect is close to that common-delay model. If this shift is independent of the data-dependent receiver timing, their variances add. The recorded comparison is 7.52703 versus 7.53176 ps^2, residual +0.00472376 ps^2. The full channel plus threshold operation and histogram estimator do not make exact equality a universal identity.

3. Does Rx case 1 variance plus Tx case 3 variance equal Rx case 3 variance?

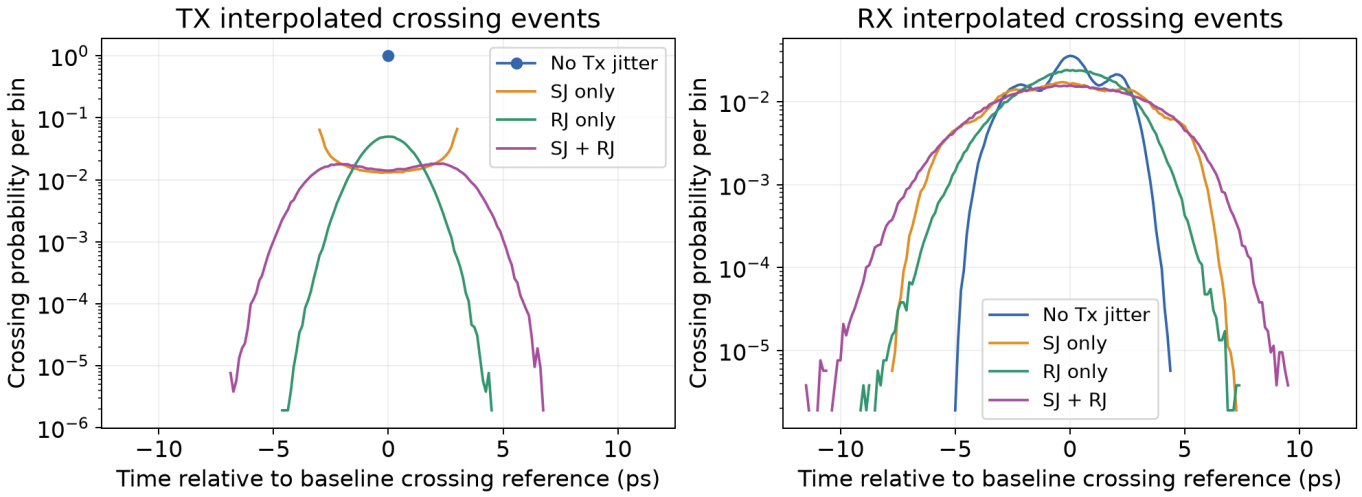
There is no general equality. Independent edge-to-edge RJ changes the widths and relative positions of neighboring pulses. Their overlapping channel responses alter both the crossing voltage slope and the crossing time. Receiver timing therefore cannot generally be represented as the baseline DDJ plus an unchanged copy of the transmitter RJ. The recorded comparison is 4.00563 versus 4.28005 ps^2, residual +0.274415 ps^2. This is not a failure of the variance-of-a-sum theorem; the proposed decomposition is the issue.

Combined jitter and measurement interpretation

4. Printed equation and the intended receiver comparison

The supplied question prints `Var(Rx1)+Var(Tx4)=Var(Tx4)`. Taken literally, the right-hand side cancels the second term and the equation requires zero baseline DDJ. The measured `Var(Rx1)=3.0074 ps^2` is nonzero, so the literal equation is false. This report preserves that reading explicitly.

The likely intended right-hand side is `Var(Rx4)`, continuing questions 2 and 3. For that interpretation, the prediction is 8.51258 ps^2 and the measured receiver case 4 value is 8.77041 ps^2, residual +0.257835 ps^2. There is no general equality for the same reason as question 3: the fast RJ is transformed together with the data pattern by the channel and the threshold crossing operation. Approximate addition of slow SJ does not make the entire combined transmitter distribution pass through unchanged.



Independent crossing-event diagnostic. Interpolated threshold crossings count edge events, while an eye slice counts time spent in a narrow voltage bin.

The threshold-slice method follows the assignment and lecture. A narrow voltage bin weights a transition approximately in proportion to the inverse of its local slope. Receiver edge slopes depend on the data pattern, so its normalized eye slice need not exactly match the histogram of individual threshold-crossing events. The additional event-based diagnostic exposes this estimator difference; it is not substituted silently for the requested eye-slice result.

Gaussian RJ is theoretically unbounded. The observed maximum and minimum in a finite run are not intrinsic peak-to-peak RJ. With about one million bits, the available low-probability evidence is around the finite-count scale; no BER extrapolation to 1e-12 or smaller is claimed. The optional ADS ChannelSim exercise has not been performed, and the assignment's example results are not presented as fresh ADS verification of this run.

Verification, corrections, and reproduction

The recorded independent suite completed **20 tests**, with 0 failures and 0 errors. It checks the circuit against an independent formulation, waveform boundary handling, block convolution against a full calculation, histogram normalization, circular phase selection and statistical calculations. Detailed results are kept in `results/test_results.json` and the verification source under `tests/`.

Recorded numerical check	Value
exact evaluated bits	True
finite outputs	True
pmfs sum to one	True
pdfs integrate to one	True
zero probabilities preserved	True

The saved-artifact verifier independently checked 16 distributions, including the time grid, all PMF/PDF normalizations and moments, exact evaluated bit counts and regenerated requested/applied jitter. The largest relative transmitter-variance difference from ideal theory is 0.4362% in this recorded run.

A separate end-to-end check evaluated 4,096 bits using block sizes 1024 and 997. All four cases produced exactly equal integer Tx/Rx eye and crossing-count arrays; only floating-point moment sums use roundoff tolerances. This checks boundary handling independently of the full-run block choice. Evidence: `results/block_size_check.json`.

The rewritten solution removes hard-coded machine paths and an unavailable legacy helper, avoids retaining all full-length waveforms, excludes startup and padded samples, and uses one consistent time spacing. It preserves zero analysis probabilities, computes real-valued thresholds, and treats PMF and PDF units separately. Every eye uses its own defined voltage grid. The source, figures, tables, and this report contain no student identity.

The calculation is a finite-bandwidth, windowed and truncated channel model. The FIR and numerical bin choices are recorded so their effects can be assessed. Independent numerical agreement validates the stated implementation; it is not an independent hardware measurement or a new circuit-simulator result.

Reproduce from the repository root

```
python -m pip install -r homework/hw12/requirements.txt
python homework/hw12/src/hw12.py
python -m unittest discover -s homework/hw12/tests -v
python homework/hw12/tests/verify_results.py --output homework/hw12/results/verification.json
python -m pip install -r homework/hw12/report/requirements.txt
python homework/hw12/report/build_report.py --refresh
```

Sources and provenance

[1] Supplied HW12 assignment, `instructions/HW12_jitter_simulation.pdf`, including the circuit, four experiments, questions and example answer. [2] Supplied `instructions/HW12_channel.s2p`. [3] Course handout 12, Jitter Basics, pages 2-12, for timing error, SJ/RJ distributions and independent-source variance addition. [4] Course handout 13, Jitter Basics (2), page 2, for the statistical-eye slice method. The instruction template is retained as a reference, not the executable solution. The independently regenerated report replaces personally identifying student reports; it does not claim that their original results were reverified.